2023 Ch H2 Q5

Section: Nature’s Chemistry

Topic: Alcohols, Esters, Natural Products

Question Summary

(a)(i) Fermentation of glucose produces ethanol and CO_2. Calculate the CO_2 volume for a scaled-up solution and determine the atom economy for ethanol.
(a)(ii) Use hydrometer readings to calculate % alcohol by volume.
(b)(i) Distillation produces fractions: first contains toxic methanol and propan-2-one. Describe chemical tests to distinguish them.
(b)(ii) Structures of plant compounds are given (limonene, myrcene, geranyl acetate). Identify compound class, isoprene unit, and hydrolysis product.
(c) Calculate the daily quinine requirement for a 70 kg adult (10 mg/kg every 8 hours).

Worked Solution

According to the SQA Marking Instructions for 2023 H2 Q5:

(a)(i)(A) Fermentation: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$.
Moles glucose in 50.0 cm^3 = 5.79 g ÷ 180 = 0.0322 mol. In 16 L solution (320 × larger), moles glucose = 10.3 mol. Moles CO_2 = 2 × 10.3 = 20.6 mol. Volume = 20.6 × 24 = 494 L CO_2.
(B) Atom economy = (mass desired product ÷ total mass products) × 100. Desired = 2 × 46 = 92, Total = 92 + 88 = 180. Atom economy = (92 ÷ 180) × 100 = 51.1%.

(a)(ii) Hydrometer readings: Initial SG = 1.080, Final SG = 1.010. Change = 0.070. % alcohol = (0.070 ÷ 0.080) × 100 = 87.5%. (MI accepts values from correct substitution).

(b)(i) Distinguishing methanol vs propan-2-one:
• Oxidation with acidified dichromate: methanol → green (oxidised to methanoic

acid); propan-2-one shows no change.
• 2,4-DNPH test: propan-2-one gives orange precipitate; methanol no reaction.

(b)(ii)(A) Limonene and myrcene are terpenes (unsaturated hydrocarbons built from isoprene units).
(B) Circle = isoprene unit (C_5H_8) within limonene.
(C) Hydrolysis of geranyl acetate produces geraniol (alcohol) and ethanoic acid.

(c) Quinine dosage: 10 mg/kg × 70 kg = 700 mg per dose. 3 doses per day (every 8 h) = 700 × 3 = 2100 mg = 2.10 g per day.

Final Answer

(a)(i)(A) CO_2 volume = 494 L. (B) Atom economy = 51.1%.

(a)(ii) % alcohol by volume = 87.5%.

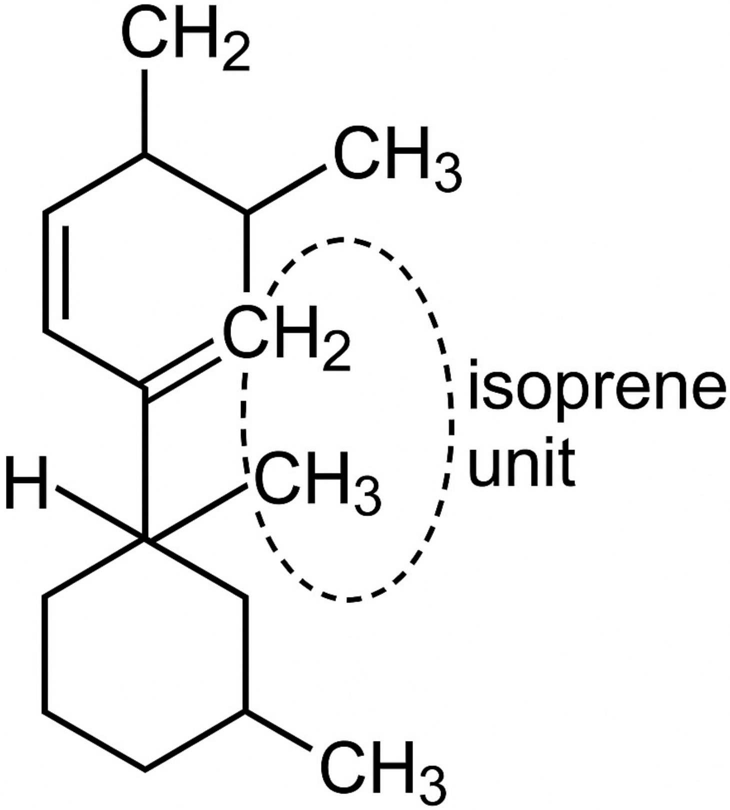
(b)(i) Methanol oxidises with dichromate; propan-2-one gives orange ppt with 2,4-DNPH.

(b)(ii)(A) Terpenes. (B) Isoprene unit circled. (C) Ethanoic acid.

(c) 2.10 g quinine per day for 70 kg adult.

Revision Tips

• Fermentation is low atom economy (only ~51% ethanol). • Hydrometer measures SG before/after fermentation → % alcohol. • Distinguish alcohols (oxidation) from ketones (2,4-DNPH). • Terpenes are built from isoprene (C_5H_8). • Quinine dosage = mg/kg × mass × number of doses.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CH3
CH2
isoprene
unit
H
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